Neurotransmitters, Neuromodulators, and Neuroendocrine Signalling

Chapter 6

In this chapter

Learning objectives 83
Introduction 83
6.1 Glutamate signalling 84
6.2 GABA inhibition 84
6.3 Noradrenergic arousal 85
6.4 Serotonergic modulation 85
6.5 Dopamine and salience 86
6.6 Acetylcholine and attention 86
6.7 Endogenous opioids 87
6.8 Endocannabinoid regulation 87
6.9 Neuropeptides 87
6.10 Hormone–brain communication 88
Synthesis 88
Composite case study 89
Review and discussion questions 90
Research assignments 90
Chapter summary 90
References 91

Central question. How do chemical signalling systems coordinate threat responses across the brain and body, and why does the same molecule produce different effects in different contexts?

Learning objectives

- distinguish neurotransmitters, neuromodulators and hormones;
- describe the principal transmitter systems relevant to threat and regulation;
- explain why "too much" or "too little" of a transmitter is rarely a sufficient explanation;
- relate neuroendocrine signalling to autonomic and behavioural responses;
- evaluate popular claims about "chemical imbalance" in trauma.

Introduction

The brain does not have a separate system for trauma. It uses the same transmitters, modulators and hormones that support movement, appetite, sleep, reward and reproduction. What changes in traumatic stress is the pattern of their engagement: which systems are recruited, for how long, and in what context.

This chapter surveys the principal signalling systems. It is organised around function rather than alphabet. Glutamate and GABA provide fast excitation and inhibition. Noradrenaline, dopamine, serotonin and acetylcholine modulate widely distributed circuits. Opioids and endocannabinoids regulate pain, reward and stress coping. Neuropeptides, including corticotropin-releasing hormone, oxytocin and neuropeptide Y, coordinate social and defensive responses. Hormones—glucocorticoids, catecholamines and gonadal steroids—carry signals between brain and body.

A caution runs through the chapter. Popular accounts often translate complex findings into a "chemical imbalance" story: too much cortisol, too little serotonin, an overactive noradrenergic system. These stories are attractive because they are simple and seem to reduce blame. They are also inaccurate. Signalling systems are nonlinear, context-dependent and reciprocal, and no single measurement in a living person reliably indexes them.

> **A note on names.** *A neurotransmitter is released at a synapse and acts locally. A neuromodulator alters the responsiveness of circuits over a wider*

area and a longer time. A hormone is released into the bloodstream and acts on distant targets. Many molecules act in more than one mode; noradrenaline, for example, is a neurotransmitter in the brainstem, a modulator in the cortex, and a hormone when released from the adrenal medulla.

6.1 Glutamate signalling

Glutamate is the principal excitatory neurotransmitter in the central nervous system. It acts at ionotropic AMPA, NMDA and kainate receptors and at metabotropic receptors. Fast excitatory transmission is mediated mainly by AMPA receptors; NMDA receptors contribute to plasticity and to slower calcium-dependent signalling; metabotropic receptors modulate release and excitability.

Glutamate is also potentially toxic. Excess extracellular glutamate overstimulates receptors, allows calcium to flood cells, and triggers cascades that can damage or kill neurons. This is why glutamate uptake by astrocytes is essential, and why conditions that impair energy supply—ischaemia, hypoglycaemia, prolonged seizures—can produce excitotoxic injury.

In trauma, glutamatergic plasticity underlies fear learning. Strengthening of thalamic and cortical inputs to the lateral amygdala, and of hippocampal and prefrontal connections, depends on NMDA receptor-dependent plasticity. This is the cellular basis of conditioned fear, and it explains why extinction is a form of new learning rather than erasure.

6.2 GABA inhibition

Gamma-aminobutyric acid is the principal inhibitory neurotransmitter. GABA-A receptors are chloride channels that produce fast inhibition; GABA-B receptors are metabotropic and produce slower, longer-lasting inhibition. Inhibitory interneurons shape the timing and synchrony of principal cell firing, and their dysfunction is implicated in epilepsy, anxiety and schizophrenia.

GABAergic function is relevant to trauma in several ways. Benzodiazepines, which enhance GABA-A receptor activity, reduce acute anxiety but are not recommended as routine treatment for PTSD because they do not address the underlying learning and may interfere with extinction. Alcohol acts on GABA-A receptors, which partly explains its use as self-medication and its withdrawal syndrome.

Inhibitory interneurons also gate fear expression. In rodents, parvalbumin-expressing interneurons in the prelimbic cortex regulate fear responses, and intercalated cells in the amygdala gate information between basolateral and central nuclei. These findings suggest that modulating inhibition, not only excitation, could change fear behaviour.

6.3 Noradrenergic arousal

Noradrenaline is produced mainly by the locus coeruleus, a small brainstem nucleus that projects throughout the forebrain. It increases arousal, alertness and vigilance, and it modulates attention, memory and stress responses. During acute stress, noradrenaline release sharpens detection of threat and supports the encoding of emotionally significant events.

Noradrenaline also has costs. High levels impair prefrontal working memory through alpha-1 receptor signalling (Arnsten, 2009), while strengthening amygdala-dependent fear learning. This is a neurochemical account of why people make worse decisions under acute threat and why traumatic memories are often vivid and fragmentary.

In PTSD, noradrenergic activity has been reported to be elevated, and drugs that reduce noradrenergic signalling—such as alpha-1 antagonists and beta-blockers—have been studied for nightmares and hyperarousal. Propranolol has been investigated for prevention after trauma and for reconsolidation-based intervention, but the evidence is not sufficient to support routine use.

6.4 Serotonergic modulation

Serotonin is produced by raphe nuclei and projects widely. It modulates mood, sleep, appetite, sexual function, pain and impulse control. Selective serotonin reuptake inhibitors are first-line pharmacotherapy for PTSD in several guidelines, with modest effect sizes.

The relationship between serotonin and trauma is not a simple deficit. Acute stress increases serotonin turnover in some regions, and chronic stress alters receptor sensitivity. Genetic variation in the serotonin transporter has been studied extensively, but candidate-gene findings have not replicated reliably in large samples, and claims that a single genotype determines trauma response should be rejected.

Serotonergic drugs also carry risks: initial anxiety, sexual dysfunction, discontinuation effects, and, in young people, increased suicidal

ideation during early treatment. These are not arguments against use; they are arguments for informed consent and monitoring.

6.5 Dopamine and salience

Dopamine is produced by the ventral tegmental area and substantia nigra and projects to striatum, prefrontal cortex and limbic regions. It signals reward prediction error, motivates approach, and modulates salience and effort. Dopamine also influences working memory in an inverted-U fashion: moderate levels optimise performance, while high or low levels impair it.

Chronic stress alters dopamine function. In animal models, stress increases dopamine release in prefrontal cortex and alters D1 receptor signalling, contributing to working-memory impairment. In humans, imaging studies report altered striatal responses to reward in PTSD, consistent with the anhedonia and diminished motivation that many patients describe.

Dopamine's role in salience makes it relevant to hypervigilance: a system that assigns excessive salience to threat-related cues will capture attention repeatedly. This is not a "dopamine excess" in the simplistic sense; it is a change in the weighting of cues within a circuit.

6.6 Acetylcholine and attention

Acetylcholine is produced by basal forebrain and brainstem nuclei and acts at nicotinic and muscarinic receptors. It supports attention, learning and memory, and it modulates cortical excitability and plasticity. Cholinergic signalling in the hippocampus and cortex is important for encoding new information.

Stress can alter cholinergic function. Acute stress increases acetylcholine release in some regions, which may support vigilance, while chronic stress is associated with cholinergic changes that impair memory. Anticholinergic drugs, common in older adults, can cause confusion and memory complaints that are sometimes mistaken for cognitive decline.

Nicotine acts at nicotinic acetylcholine receptors and is highly addictive. Smoking is common in people with trauma histories, partly because nicotine temporarily reduces anxiety and improves attention. Treating smoking as a moral failure rather than a dependence is both ineffective and unjust.

6.7 Endogenous opioids

Endogenous opioids include endorphins, enkephalins, dynorphins and endomorphins. They act at mu, delta and kappa receptors and modulate pain, reward, stress and social bonding. Endogenous opioid release can produce analgesia during injury and a sense of relief after danger.

Opioid signalling is implicated in dissociation and numbing. Animal studies suggest that kappa opioid receptor activation can produce aversive, dysphoric states and social withdrawal, while mu receptor activation produces analgesia and euphoria. In humans, opioid blockade has been used experimentally to study the role of endogenous opioids in stress responses.

The clinical implications are complicated. Prescription opioids carry substantial risks and are not indicated for PTSD. Buprenorphine, a partial mu agonist and kappa antagonist, has been studied for treatment-resistant depression and, more recently, for PTSD, but the evidence is preliminary.

6.8 Endocannabinoid regulation

Endocannabinoids—anandamide and 2-arachidonoylglycerol—are synthesised on demand and act at CB1 receptors, which are abundant in cortex, hippocampus, amygdala and striatum. They regulate neurotransmitter release, modulate stress responses, and support fear extinction.

CB1 receptor activation in the amygdala and prefrontal cortex facilitates extinction learning and reduces anxiety-like behaviour in animal models. In humans, cannabis use is common among people with PTSD, often for sleep and anxiety. However, cannabis use is also associated with worse outcomes in some studies, and the evidence for cannabinoids as treatment is limited and mixed.

The endocannabinoid system illustrates a general problem: a plausible mechanism does not make an intervention effective or safe. Clinical decisions require trials, not mechanistic enthusiasm.

6.9 Neuropeptides

Neuropeptides are short chains of amino acids that act as neurotransmitters or modulators. Corticotropin-releasing hormone (CRH) drives the HPA axis and also acts centrally to increase anxiety.

Oxytocin is involved in social bonding, childbirth and lactation, and has context-dependent effects on trust and defence. Neuropeptide Y is associated with stress resilience. Substance P and calcitonin gene-related peptide are involved in pain signalling.

CRH is a central target in trauma research. CRH receptor antagonists have been studied for depression and anxiety, with mixed results. Oxytocin has been studied for PTSD with small and inconsistent effects. Neuropeptide Y has attracted interest because higher levels have been associated with resilience in some studies, including in military populations.

These systems are promising but immature. Clinical claims should be proportionate to the evidence, and "natural" peptides are not automatically safe.

For an overview of stress neurobiology, including transmitter and neuropeptide systems, see Godoy et al. (2018).

6.10 Hormone–brain communication

Hormones reach the brain through the bloodstream and act on receptors in the hypothalamus, hippocampus, amygdala and prefrontal cortex. Glucocorticoids (cortisol in humans, corticosterone in rodents) are released from the adrenal cortex under HPA-axis control. They mobilise energy, modulate immune function, and shape memory.

Glucocorticoids have inverted-U effects on memory: moderate levels support consolidation, while very high or very low levels impair it. They also exert rapid non-genomic effects through membrane receptors, distinct from slower genomic effects. This helps explain why the same hormone can produce different effects at different timescales.

Gonadal steroids—oestrogen, progesterone and testosterone—modulate stress reactivity, fear learning and extinction. Oestrogen, for example, appears to enhance extinction recall in some studies, which may contribute to sex differences in PTSD prevalence. These findings are important and under-studied.

The endocrine detail is reviewed by Herman et al. (2016).

Synthesis

Table 6.1 summarises the principal signalling systems.

Table 6.1. Signalling systems and trauma-relevant functions

System	Main source	Principal functions	Trauma relevance	Cautions
Glutamate	Cortical and subcortical neurons	Fast excitation, plasticity	Fear learning, excitotoxicity	Context-dependent
GABA	Interneurons	Fast inhibition, network timing	Fear gating, anxiety	Benzodiazepines not first-line for PTSD
Noradrenaline	Locus coeruleus	Arousal, vigilance, attention	Hyperarousal, intrusive memory	Inverted-U effects
Serotonin	Raphe nuclei	Mood, sleep, impulse control	PTSD pharmacotherapy	Effect sizes modest
Dopamine	VTA, substantia nigra	Salience, reward, effort	Anhedonia, hypervigilance	Nonlinear dose effects
Acetylcholine	Basal forebrain, brainstem	Attention, encoding	Memory complaints	Measured indirectly
Opioids	Distributed	Analgesia, reward, numbing	Dissociation, self-medication	Dependence risk
Endocannabinoids	On demand	Modulate release, extinction	Sleep, anxiety, extinction	Evidence mixed
CRH	Hypothalamus, extended amygdala	HPA drive, anxiety	Stress response	Trials disappointing
Glucocorticoids	Adrenal cortex	Energy, immunity, memory	Consolidation, feedback	Inverted-U effects

Composite case study

A patient reports that alcohol is the only thing that helps him sleep. A neurotransmitter-informed formulation recognises that alcohol enhances GABA-A signalling and temporarily reduces noradrenergic arousal, which explains the relief and the rebound insomnia. It does not conclude that he has a GABA deficiency. The plan addresses sleep,

trauma reminders and alcohol withdrawal, with medication reviewed for interactions.

Review and discussion questions

Why is "chemical imbalance" a poor explanation of trauma-related distress?

How do the effects of noradrenaline on prefrontal cortex and amygdala differ, and why does this matter clinically?

What is the evidence for and against benzodiazepines in PTSD?

Why are candidate-gene findings in psychiatry often unreliable?

How might endocannabinoid signalling explain why some people use cannabis after trauma, and what are the risks?

Why is it difficult to measure neurotransmitter function in living humans?

Research assignments

- Design a study of sex differences in extinction recall, specifying hormonal assessment and cycle phase.
- Critically appraise the evidence for a CRH antagonist in PTSD.
- Write a patient information leaflet on SSRIs for PTSD that is accurate about benefits and adverse effects.

Chapter summary

Neurotransmitters and hormones coordinate threat responses across brain and body. Glutamate and GABA provide fast excitation and inhibition; noradrenaline, dopamine, serotonin and acetylcholine modulate attention, salience, mood and memory; opioids and endocannabinoids regulate pain, reward and coping; neuropeptides and hormones link brain and body. Effects are nonlinear and context-dependent, and no single chemical measure identifies trauma. Popular "imbalance" stories should be replaced by accurate accounts of systems, contexts and uncertainty.

References

Arnsten, A. F. T. (2009). Stress signalling pathways that impair prefrontal cortex structure and function. *Nature Reviews Neuroscience, 10*(6), 410–422. https://doi.org/10.1038/nrn2648

Godoy, L. D., Rossignoli, M. T., Delfino-Pereira, P., Garcia-Cairasco, N., & de Lima Umeoka, E. H. (2018). A comprehensive overview on stress neurobiology: Basic concepts and clinical implications. *Frontiers in Behavioral Neuroscience, 12*, Article 127. https://doi.org/10.3389/fnbeh.2018.00127

Herman, J. P., McKlveen, J. M., Ghosal, S., Kopp, B., Wulsin, A., Makinson, R., Scheimann, J., & Myers, B. (2016). Regulation of the hypothalamic–pituitary–adrenocortical stress response. *Comprehensive Physiology, 6*(2), 603–621. https://doi.org/10.1002/cphy.c150015